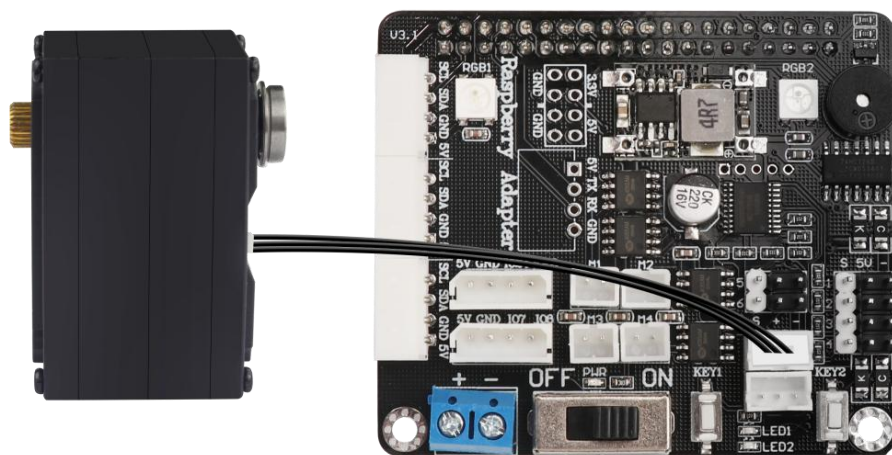


Lesson 2 Read Bus Servo Status

1. Preparation

1.1 Hardware wiring

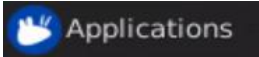
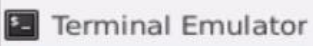
Connect bus servo to any bus servo port of Raspberry Pi board separately. Here we take LX-15D as an example.

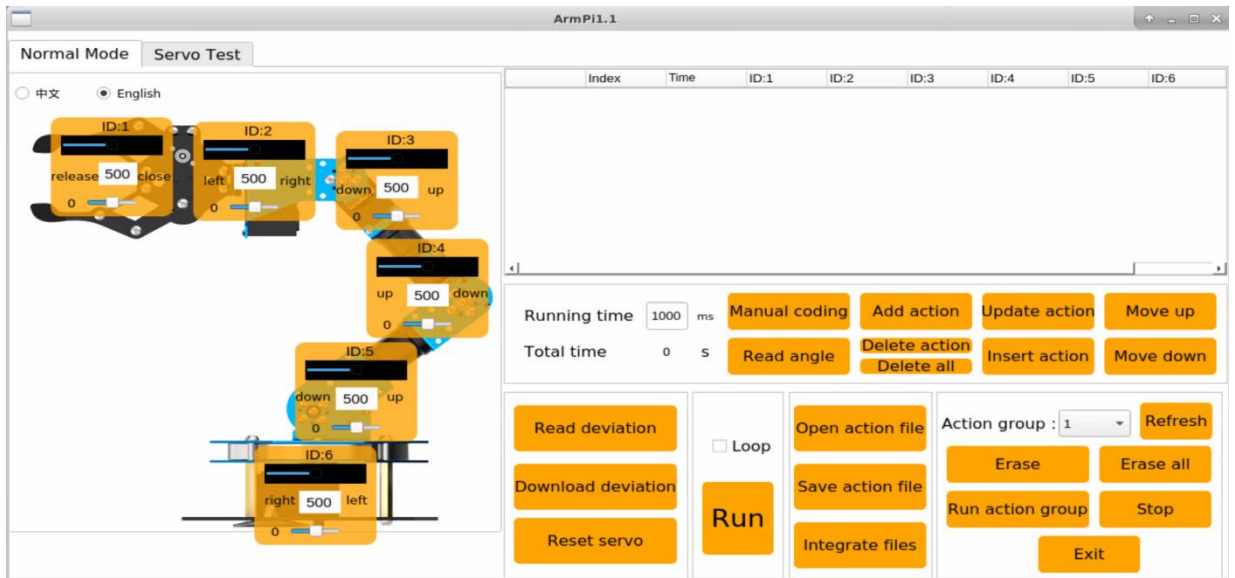


Note: the bus servo wire adopts anti-reverse plug design, please insert carefully.

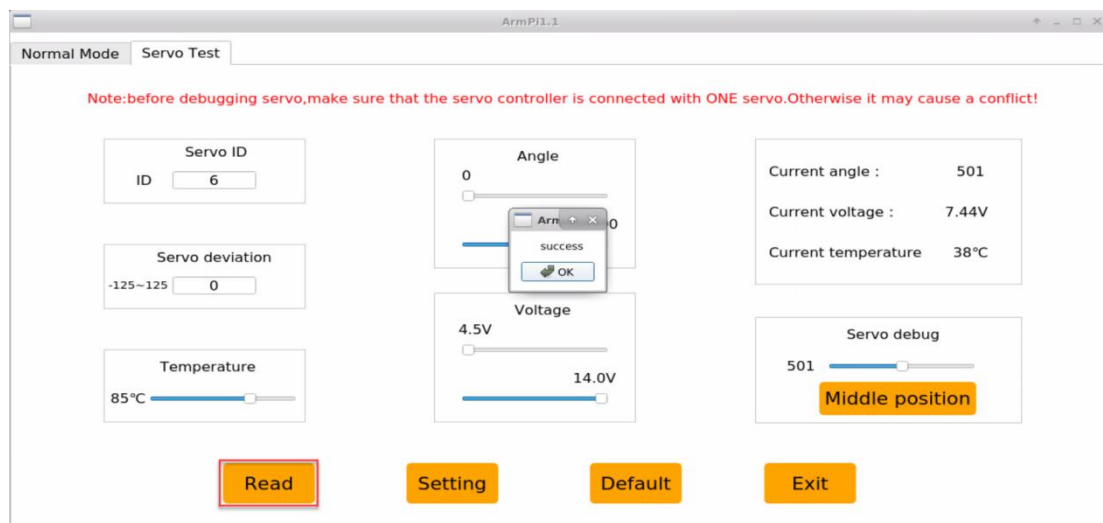
1.2 ID Set

The sample program default reads the ID 6 servo status information (position, temperature, voltage). It can be set through the "Servo Test" in the ArmPi FPV PC computer.

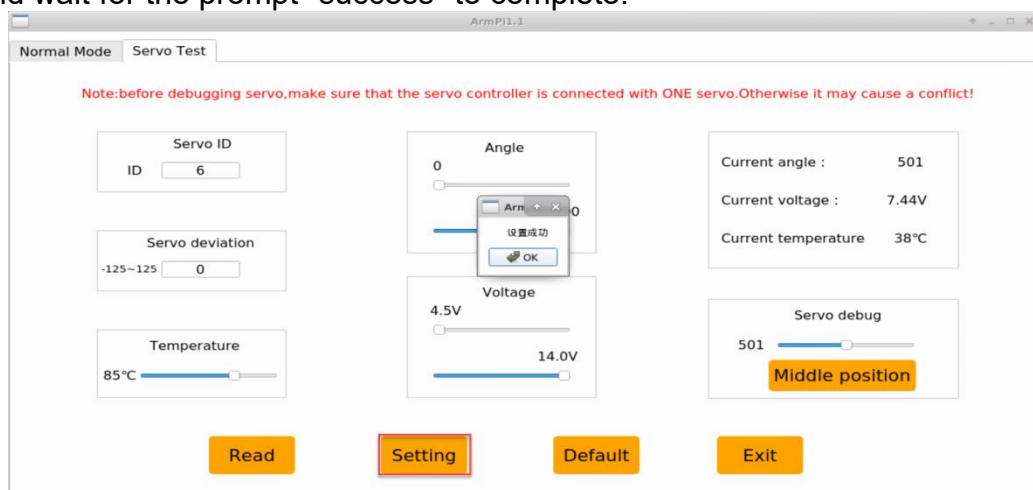
- 1) Open LX terminal. Click  and select  to open command line terminal of Ubuntu. Lastly, enter "sudo ArmPi_PC_Software/ArmPi.py" command in terminal window to open PC software.



- 2) Click and switch to the "Servo Test" menu bar, click "Read" button, and wait for the prompt "success".



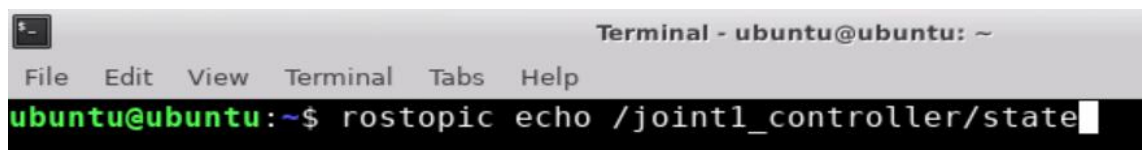
- 3) In the "Servo ID" area, select the ID, input servo number 6, click "Set" button, and wait for the prompt "success" to complete.



2. Operation Steps

1) Open LX terminal. Click  Applications and select  Terminal Emulator to open command line terminal of Ubuntu.

2) Enter “**rostopic echo /joint1_controller/state**” command and press Enter.

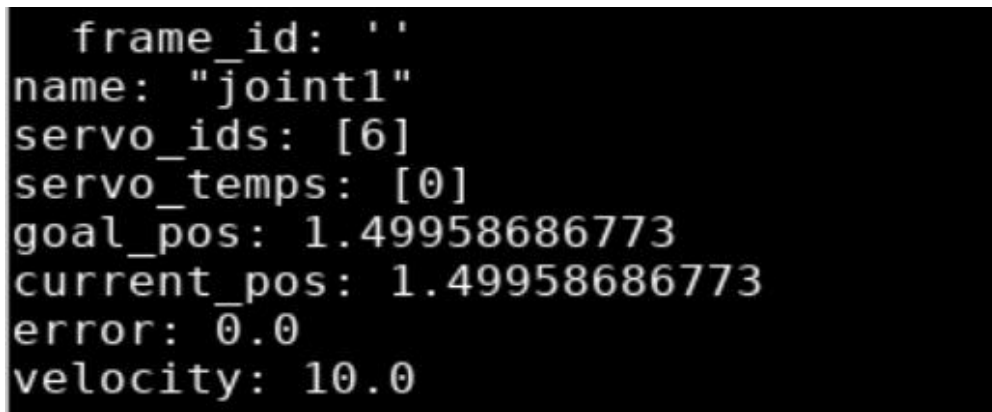


```
Terminal - ubuntu@ubuntu: ~  
File Edit View Terminal Tabs Help  
ubuntu@ubuntu:~$ rostopic echo /joint1_controller/state
```

3) If you want to quit this program, press “Ctrl+C”

3. Project Outcome

After the command is confirmed, the name, ID and location of servo will be listed in the interface.



```
frame_id: ''  
name: "joint1"  
servo_ids: [6]  
servo_temps: [0]  
goal_pos: 1.49958686773  
current_pos: 1.49958686773  
error: 0.0  
velocity: 10.0
```

name: servo name

servo_ids: ID

goal_ops: target position

current_ops: current position